EASY · PROBLEM-SOLVING AND DATA ANALYSIS · ANSWER KEY

Problem-Solving and Data Analysis

14

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 A

A. 5

Choice A is correct. It’s given that the ratio of black pens to red pens is 8 to 1. Therefore, there are $\frac{1}{8}$ as many red pens as black pens in the box. It’s also given that there are 40 black pens in the box. Therefore, the number of red pens is $\frac{1}{8}$ of the 40 black pens. Thus, the number of red pens is $40\frac{1}{8}$, or 5.

Choice B is incorrect. This is the number of black pens in the box for every red pen.

Choice C is incorrect. This is the number of black pens in the box.

Choice D is incorrect. This is the number of red pens in the box if the ratio of black pens to red pens is 1 to 8.

Q2 17

The correct answer is 17. It's given that there are 170 blocks in a container, and of these blocks, 10% are green. Since 10% can be rewritten as $\frac{10}{100}$, or 0.1, the number of green blocks in the container is 0.1170, or 17.

Q3 B

B. 4.2

Choice B is correct. The numbers of defective lightbulbs found for the five trials are 4, 7, 1, 3, and 6, respectively. The mean is therefore $\frac{4+7+1+3+6}{5}=4.2$.

Choice A is incorrect. This is the median number of defective lightbulbs for the five trials. Choice C is incorrect and may result from an arithmetic error. Choice D is incorrect and may result from mistaking the number of trials for the number of defective lightbulbs.

Q4 B

B. $y = 13.5 - 0.8x$

Choice B is correct. The line of best fit shown intersects the y-axis at a positive y-value and has a negative slope. The graph of an equation of the form $y = a + bx$, where a and b are constants, intersects the y-axis at a y-value of a and has a slope of b. Of the given choices, only choice B represents a line that intersects the y-axis at a positive y-value, 13.5, and has a negative slope, -0.8.

Choice A is incorrect. This equation represents a line that has a positive slope, not a negative slope.

Choice C is incorrect. This equation represents a line that intersects the y-axis at a negative y-value, not a positive y-value, and has a positive slope, not a negative slope.

Choice D is incorrect. This equation represents a line that intersects the y-axis at a negative y-value, not a positive y-value.

Q5 A

A. 17

Choice A is correct. It's given that the ratio of the length of line segment XY to the length of line segment ZV is 6 to 1, which means $\frac{XY}{ZV} = \frac{6}{1}$. It's given that the length of line segment XY is 102 inches. If the length, in inches, of line segment ZV is represented by l, the value of l can be calculated by solving the equation $\frac{102}{l} = \frac{6}{1}$, or $\frac{102}{l} = 6$. Multiplying each side of this equation by l yields $102 = 6l$. Dividing each side of this equation by 6 yields $17 = l$. Therefore, the length of line segment ZV is 17 inches.

Choice B is incorrect. This is the length, in inches, of line segment ZV if the length of line segment XY is 576, not 102, inches.

Choice C is incorrect. This is the length, in inches, of line segment XY, not line segment ZV.

Choice D is incorrect. This is the length, in inches, of line segment ZV if the ratio of the length of line segment XY to the length of line segment ZV is 1 to 6, not 6 to 1.

Q6

B

B. 37

Choice B is correct. 10% of a quantity means $\frac{10}{100}$ times the quantity. Therefore, 10% of 370 can be represented as $\frac{10}{100}370$, which is equivalent to 0.10370, or 37. Therefore, 10% of 370 is 37.

Choice A is incorrect. This is 10% of 270, not 10% of 370.

Choice C is incorrect. This is 90% of 370, not 10% of 370.

Choice D is incorrect. This is $370 - 10$, not 10% of 370.

Q7

A

A. 10

Choice A is correct. It's given that the bar graph shows the distribution of the number of students in each of four extracurricular activities at a high school. The bar representing drama has a height of 40; therefore, there are 40 students in drama. The bar representing chess has a height of 30; therefore, there are 30 students in chess. Thus, there are 40 - 30, or 10 more students in drama than in chess.

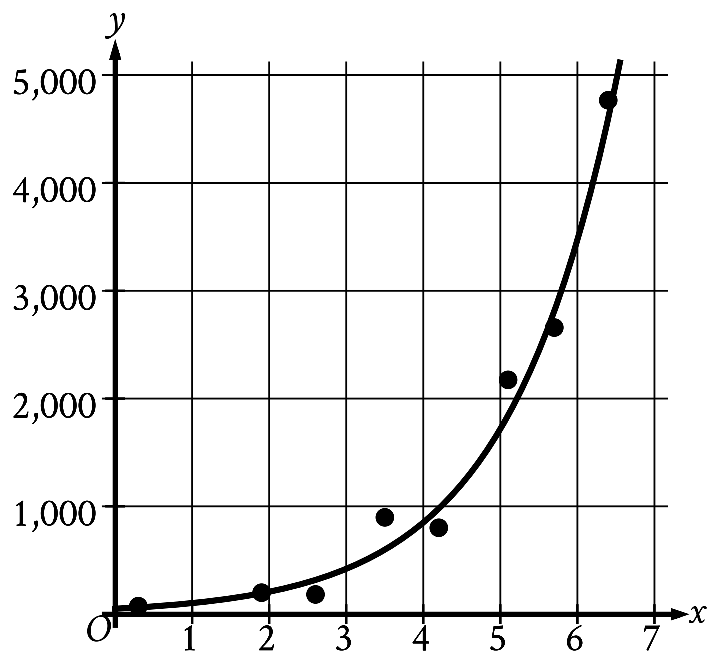
Choice B is incorrect. This is the number of students in chess.

Choice C is incorrect. This is the number of students in drama.

Choice D is incorrect. This is the sum of the number of students in drama and in chess.

Q8

D



D.

- The scatterplot has 8 data points.
- The data points are in an exponential pattern trending up from left to right.
- A curve of best fit is shown:
  - The curve of best fit trends up from left to right.
  - 2 points are touching the curve of best fit.
  - 3 points are above the curve of best fit.
  - 3 points are below the curve of best fit.
  - The curve of best fit passes through the following approximate coordinates:
    - (0 comma 51)
    - (4 comma 851)
    - (6 comma 3,472)

Choice D is correct. An appropriate model should follow the trend of the data points and should have data points both above and below the model. The scatterplot shows that the data points have an increasing trend that is curved. Therefore, an appropriate model should be an increasing curve with data points both above and below the model. Of the given choices, only the model in choice D is an increasing curve with data points both above and below the model.

Choice A is incorrect. Since the trend of the data points isn't linear, a line isn't the most appropriate model for the data.

Choice B is incorrect. Since the trend of the data points is increasing and isn't linear, a decreasing line isn't the most appropriate model for the data.

Choice C is incorrect. All the data points are below the model shown in this graph.

Q9 66

The correct answer is 66. It’s given that a product costs 11.00 dollars per pound. Therefore, the cost for 6 pounds of the product is $\frac{11.00 \text{ dollars}}{1 \text{ pound}} 6$ pounds, which is equivalent to 66.00, or 66, dollars.

Q10 D

D. 100

Choice D is correct. Let x represent the number that 21 is 21% of. It follows that $\frac{21}{x} = \frac{21}{100}$. Multiplying each side of this equation by x yields $21 = \frac{21x}{100}$. Multiplying each side of this equation by 100 yields $2{,}100 = 21x$. Dividing each side of this equation by 21 yields $100 = x$. Therefore, 21 is 21% of 100.

Choice A is incorrect. 21% of 0 is 0, not 21.

Choice B is incorrect. 21% of 1 is 0.21, not 21.

Choice C is incorrect. 21% of 42 is 8.82, not 21.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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